

A Bayesian Framework for Causal Inference with Error-Prone Environmental Exposures

Johnson Vo

Supervisor: Dr. Kuan Liu

Committee: Dr. Olli Saarela, Dr. Hong Chen



Dalla Lana School of Public Health
The University of Toronto
March 2026

A proposal submitted to the University of Toronto in partial fulfillment of the requirements of
the Masters' of Science Thesis Program

©Johnson Vo, 2026

Key Words

Bayesian Causal Inference, Air Pollution

1 Introduction

Air pollution remains a leading environmental risk factor for public health globally, contributing to morbidity and mortality in millions of people each year¹. In Canada, climate change is increasing the frequency and intensity of extreme environmental events, with recent wildfire seasons producing unprecedented levels of smoke-related PM_{2.5} exposure across large geographic regions^{2,3}. These exposures have been linked to a wide range of adverse health outcomes, including mortality and cardiorespiratory disease, and are expected to play an increasingly important role in population health as wildfire activity intensifies⁴. Large-scale environmental studies investigating the health effects of PM_{2.5} increasingly rely on model-based exposure estimates rather than dense networks of ground monitoring stations. Satellite data and machine learning models are commonly used to generate high-resolution exposure surfaces that estimate pollutant concentrations over large geographic areas⁵. In Canada, nationally harmonized exposure infrastructure (e.g., CANUE⁶) and widely used retrospective smoke PM_{2.5} surfaces (e.g., CanOSSEM⁵) substantially lower barriers to population-level studies by providing analysis-ready exposures linked to health records at residential postal codes. These resources have substantially expanded the scale and scope of environmental health studies.

Despite these advances, the use of model-based exposure products introduces several important challenges and limitations. Error-prone exposure occurs when the exposure assigned at an areal level may not match individual-level exposure⁷. The effects of error-prone exposure on health effect estimates have been well studied and documented^{8,9}. For example, simulation studies of modeled air pollution exposures have shown that measurement error can lead to significant bias in estimated health effects. A study by Samoli et al.¹⁰ reported bias ranging from -20% to +20% on mortality depending on the exposure model and study design. Consequently, the exposure variable used in statistical analyses may differ from the true environmental exposure relevant to disease risk.

Two major sources of exposure measurement error arise in air pollution studies. First, modelled surfaces are trained and validated against monitoring networks that are spatially sparse and non-randomly placed which yields systematically larger prediction errors in rural areas. When these error patterns correlate with population composition, the resulting exposure error is not “random noise” but can be differential with respect to key covariates. Secondly, even when ambient pollutant concentrations are accurately estimated at residential locations, these values may differ from the personal exposure experienced by individuals. Personal exposure depends on time-activity patterns, housing characteristics, occupational environments, and other behavioral factors that influence the relationship between ambient pollution levels and individual inhalation exposure. Research has shown that ambient-based exposure may not be uniform across populations with meaningful variation present through socio-economic status¹¹. Individuals residing within the same geographic area may experience heterogeneous exposure levels due to local environmental conditions and differences in behavior or housing. As a result, the exposure variable used in health models may systematically differ from the true environmental exposure relevant to disease risk. Treating assigned ambient exposures as fixed, error-free covariates may introduce bias and discrepancies and obscure important exposure-response relationships across populations.

Given the increasing reliance on modelled environmental exposure products in population health research, there is a need for statistical methods that explicitly account for measurement error and spatial misalignment in exposure data while supporting valid causal inference. This thesis aims to contribute to this area by developing a Bayesian framework for causal inference with error-prone environmental exposures, motivated by wildfire smoke PM_{2.5} exposure assigned at the postal-code level and applied to individual-level mental health outcomes.

2 Literature review

Early methodological work has demonstrated that measurement error in air pollution exposure estimates can substantially influence estimated health effects. Exposure estimates derived from prediction models can contain complex mixtures of different forms of measurement error that are attributed to either instrument noise or aggregation error. This can affect bias and uncertainty in health effect estimates in different ways, highlighting the importance of explicitly modelling exposure uncertainty¹². Subsequent work has further examined how prediction errors in modeled exposure surfaces arise from the spatial distribution of monitoring networks and the statistical models used to generate exposure predictions^{10,13}. These studies demonstrate that exposure prediction errors may be spatially structured and may vary systematically across geographic regions depending on factors such as monitor density, meteorology, and land use characteristics. In spatial settings, additional complications arise because exposures are spatially autocorrelated and are often misaligned with administrative health geographies (e.g., gridded exposures assigned to postal codes), leading to spatially structured error components.

A variety of statistical methods have been proposed to address exposure measurement error in epidemiologic studies. Classical approaches include regression calibration¹⁴ and simulation–extrapolation¹⁵, which attempt to correct bias introduced by measurement error either using auxiliary data or simulated error structures. Bayesian hierarchical measurement error models have also been developed to treat the true exposure as a latent variable and jointly model the exposure and outcome processes, allowing uncertainty in exposure predictions to propagate into health effect estimation^{16,17}.

More recently, causal inference methods have been increasingly adopted in environmental epidemiology to estimate the health effects of air pollution exposures. Exposure assignment in observational studies is not randomized and therefore individuals exposed to different pollution levels may differ systematically in socioeconomic characteristics, baseline health status, and subject to confounding. Methods based on the potential outcomes framework, such as g-computation¹⁸, provide a principled approach for estimating causal exposure–response relationships in observational settings. These approaches have been applied in large air pollution cohort studies to estimate the health effects of long-term PM_{2.5} exposure while accounting for confounding and exposure uncertainty¹⁹.

Wu et al.²⁰ proposes a causal framework implementing regression calibration through validated data with generalized propensity score to adjust for both causal confounding and error-prone exposure. Their method treats the true exposure as an unobserved quantity and uses auxiliary exposure models to correct bias induced by measurement error in predicted pollution levels. Further work has shown that the misclassification of treatment assignment can propagate measurement error in both the treatment and outcome model, with methods such as likelihood-based adjustment through validation data being able to adjust for this misclassification²¹. Schwartz et al.²² extends this further by introducing a space-time varying regression calibration framework applied to continuous exposure–response curves. Josey et al.²³ introduces a fully Bayesian approach that treats the true exposure as a latent variable and integrates exposure modeling directly into the causal inference procedure. This links the observed predictions to the latent exposure and accounts for uncertainty that is induced when correcting for error-prone exposure.

Additional complexity found in environmental causal inference is through confounding in spatial settings. Papadogeorgou et al.²⁴ develop distance-adjusted propensity score matching to mitigate unmeasured spatial confounding by incorporating geographic proximity into the matching process, while Papadogeorgou and Dominici²⁵ propose a Bayesian framework for estimating causal exposure–response functions with local confounding adjustment, allowing the set and influence of confounders to vary across the exposure range and adjust for spatial confounding. Together, these approaches highlight that, beyond measurement error, environmental causal inference must also contend with spatial structure and complex confounding mechanisms, motivating the need for flexible frameworks that can jointly address these challenges.

Despite this progress, several methodological gaps remain.

- **Covariate-dependent exposure error is under-modeled.** Many error-prone exposure models assume relatively simple measurement error structures that do not allow exposure uncertainty to vary

systematically with covariates (e.g., constant variance within administrative units). In practice, exposure error plausibly varies with neighbourhood in characteristics such as rurality, meteorology, land use patterns, and monitoring network placement distribution. A Bayesian framework that allows exposure error to depend on such covariates may provide a more realistic representation of exposure uncertainty. Existing work has addressed categorical causal effects after regression calibration, or continuous exposure-response estimation with multiple imputation / regression calibration, but not a fully Bayesian multilevel framework targeting intervention-specific average potential outcomes and contrasts for a latent area-level smoke exposure with individual outcomes nested within areas.

- **Limited integration with individual-level health outcomes.** Available approaches have focused largely on aggregated settings, whereas many environmental health studies involve individual-level outcomes linked to area-level exposures. Specifically, much of the existing literature on wildfire smoke exposure focuses on acute health outcomes such as mortality or cardiorespiratory events and is often implemented at aggregated spatial levels. Mental health outcomes such as incident depression remain comparatively understudied in the context of wildfire smoke exposure despite increasing evidence that wildfire events may have broader health consequences. A model that links postal-code exposure histories to *individual* incident mental health outcomes is both scientifically timely and methodologically informative.
- **Insufficient simulation evidence on equity-relevant bias patterns.** There is limited simulation evidence evaluating how spatially structured exposure error affects causal inference in environmental epidemiology. In particular, few studies consider settings that reflect real-world exposure systems—such as sparse monitoring networks, spatially correlated exposures, and prediction errors that vary systematically with factors like rurality or socioeconomic status. As a result, it remains unclear when naive analyses yield biased estimates and when proposed correction methods meaningfully improve inference.

3 Objectives

The overarching goal is to develop and evaluate a fully Bayesian causal inference method for latent postal-code-level ambient wildfire-smoke $PM_{2.5}$ exposure that is measured with error, with application to individual incident cognitive impairment.

Specifically, motivated by the limitations of existing literature, this thesis will develop a Bayesian causal inference framework for wildfire smoke $PM_{2.5}$ exposure that incorporates exposure measurement error that may vary with measured area-level covariates when exposures are assigned at the postal-code level. By explicitly modelling uncertainty in environmental exposure estimates while estimating causal exposure–response relationships, this work aims to provide improved tools for studying the health impacts of wildfire smoke in population health datasets.

Our proposed method differs from existing error-prone causal works in both target and inferential strategy. Rather than estimating a smoothed continuous exposure-response curve through a multiple-imputation procedure or through regression calibration, we target posterior distributions of average potential outcomes and causal contrasts under specified exposure interventions. Motivated by these limitations, this thesis will develop a Bayesian causal inference framework via g-computation for wildfire-smoke $PM_{2.5}$ exposure that permits measurement error in the postal-code-level surrogate to depend on measured area-level calibration covariates. Further, we will consider loosening unnecessary parametric assumptions on the covariate distribution via a non-parametric Bayesian bootstrap.

4 Methods

4.1 Notation

Let $i = 1, \dots, I$ index areal units and let $j = 1, \dots, n_i$ index individuals within area i . For individual j in area i , let Y_{ij} denote the observed outcome, let L_{ij} denote a vector of individual-level pre-exposure covariates, and let X_i denote a vector of area-level pre-exposure covariates. Let A_i denote the latent true area-level exposure and let Z_i denote the observed error-prone exposure surrogate.

Let D_i denote a vector of observed area-level calibration covariates that may explain the magnitude or direction of error in the observed exposure surrogate (e.g., season, monitor density). D_i may overlap with X_i . We distinguish them conceptually because X_i indexes confounding control whereas D_i indexes the exposure measurement error mechanism.

We write $\mathcal{D}^{\text{obs}} = \{Y_{ij}, L_{ij}, X_i, D_i, Z_i : i = 1, \dots, I, j = 1, \dots, n_i\}$ for the observed data and $\mathcal{D}^{\text{comp}} = \{Y_{ij}, L_{ij}, X_i, D_i, Z_i, A_i : i = 1, \dots, I, j = 1, \dots, n_i\}$ for the corresponding complete data, where the latent true exposures $\{A_i\}_{i=1}^I$ are unobserved.

Let W denote the known spatial structure among areal units, for example a binary adjacency matrix. We treat W as fixed and use it to define the prior dependence structure of a latent spatial random effect $\phi = (\phi_1, \dots, \phi_I)^\top$. Thus, W is not modelled as a random variable, but a known design object that indexes spatial dependence. This specification matches with the usual role of an adjacency matrix in CAR-type spatial models and with standard spatial causal formulations.

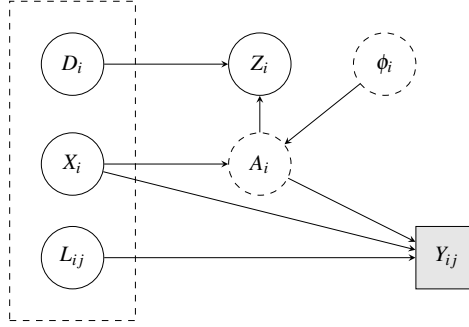


Figure 1: Causal directed acyclic graph for the proposed Bayesian causal inference framework with error-prone environmental exposure. A_i denotes the latent true postal-code-level ambient wildfire-smoke $\text{PM}_{2.5}$ exposure, Z_i the observed error-prone surrogate, X_i area-level pre-exposure confounders, D_i area-level calibration covariates for the measurement error model, ϕ_i a latent spatial random effect, and Y_{ij} the individual-level outcome for subject j in area i . Individual-level covariates L_{ij} .

4.2 Causal assumptions and estimand

We adopt the potential outcomes framework. Let $Y_{ij}(a)$ denote the potential outcome that would be observed for individual j in area i under an intervention that sets the latent true exposure in area i to level a . Our target estimand is the average potential outcome under exposure level a ,

$$\mu(a) = \mathbb{E}\{Y_{ij}(a)\}, \quad (1)$$

where the expectation is taken with respect to the target population distribution of measured individual- and area-level covariates.

Because the latent true exposure A_i is not observed, it is helpful to distinguish two levels of assumptions. First, if A_i were observed, then the causal estimand in (1) would be identified from standard causal assumptions. Second, in the actual observed data, only the error-prone surrogate Z_i is available, so inference additionally relies on assumptions about the exposure measurement error model that links Z_i to A_i .

Consistency. For each individual and area, if the latent true exposure in area i equals a , then $Y_{ij} = Y_{ij}(a)$. That is, the observed outcome equals the potential outcome under the realized latent true exposure.

Conditional exchangeability. For all exposure levels a in the support of interest, $Y_{ij}(a) \perp A_i \mid L_{ij}, X_i$. That is, conditional on the measured individual- and area-level pre-exposure covariates, the latent true exposure is assumed independent of the potential outcomes.

Positivity. For all covariate values (l, x) with positive probability in the target population, $f_{A|L,X}(a \mid l, x) > 0$ for the exposure levels a under consideration.

No interference. We assume that one area's exposure does not directly alter another area's potential outcomes beyond what is encoded through its own exposure level. Equivalently, the potential outcome for individual (i, j) is indexed by the exposure assigned to area i only. Here, spatial dependence in the observed data is represented through the latent exposure process rather than through spillover in the outcome-generating mechanism.

Under these assumptions, the average potential outcome is identified by the g-formula

$$\begin{aligned} \mu(a) &= \mathbb{E}\{Y_{ij}(a)\} \\ &= \int \mathbb{E}\{Y_{ij}(a) \mid L_{ij} = l, X_i = x\} dP(l, x) \\ &= \int \mathbb{E}\{Y_{ij}(a) \mid A_i = a, L_{ij} = l, X_i = x\} dP(l, x) \\ &= \int \mathbb{E}\{Y_{ij} \mid A_i = a, L_{ij} = l, X_i = x\} dP(l, x), \end{aligned} \tag{2}$$

where $P(l, x)$ denotes the target population distribution of the measured covariates (L_{ij}, X_i) .

In the present problem, however, A_i is not observed. Instead, we observe the surrogate exposure Z_i . Therefore, identification and estimation from the observed data additionally rely on assumptions about the exposure measurement error mechanism. In particular, the proposed Bayesian model assumes that Z_i is linked to the latent true exposure A_i through a measurement error model, and that the latent exposure process itself can be modelled using measured area-level covariates and spatial structure. Thus, causal inference for $\mu(a)$ is based not only on the standard causal assumptions above, but also on the modelling assumptions used to recover information about the unobserved exposure A_i from the observed surrogate Z_i .

Exposure measurement error. Because A_i is not observed, causal inference from the observed data further relies on assumptions about the relationship between the surrogate Z_i and the latent exposure A_i . We assume a non-differential surrogate mechanism, $Y_{ij} \perp Z_i \mid A_i, L_{ij}, X_i, D_i$, where conditioning on the latent true area-level exposure and measured covariates, the surrogate exposure carries no additional information about the outcome. Furthermore, we allow the error in Z_i to depend on D_i , and we assume that any information used to learn this relationship is transportable to the target study population. When direct calibration data are unavailable, posterior inference about A_i is necessarily more model- and prior-dependent, so the measurement error specification will be examined in sensitivity analyses.

4.3 Proposed Bayesian g-computation with error-prone environmental exposure

To avoid overloading notation, we use distinct parameter blocks for the different model components. Let η_Y , η_Z , and η_A denote the parameters of the outcome model, the exposure measurement error model, and the latent true exposure model, respectively. Let ψ_ϕ denote the hyperparameters governing the spatial prior. For compact posterior notation, we write $\Theta = (\eta_Y, \eta_Z, \eta_A, \phi, \psi_\phi)$.

We specify three linked model components: an outcome model, an exposure measurement error model, and a latent true exposure model with spatial dependence. To keep the focus on the exposure

measurement error problem, we do not impose a fully parametric model on the nuisance covariate distribution used for standardization. Instead, we estimate that distribution nonparametrically using the Bayesian bootstrap.

First, the conditional outcome model is

$$Y_{ij} | A_i, L_{ij}, X_i \sim f_Y(\cdot | A_i, L_{ij}, X_i; \eta_Y). \quad (3)$$

Second, the observed error-prone exposure is linked to the latent true exposure through

$$Z_i | A_i, D_i \sim f_Z(\cdot | A_i, D_i; \eta_Z). \quad (4)$$

Third, the latent true exposure model is

$$A_i | X_i, \phi_i \sim f_A(\cdot | X_i, \phi_i; \eta_A), \quad (5)$$

where the spatial random effect satisfies

$$\phi | W, \psi_\phi \sim f_\phi(\cdot | W; \psi_\phi). \quad (6)$$

For areal data, a common choice is a conditional autoregressive prior, with W the known adjacency matrix. In this formulation, spatial dependence is introduced through the latent exposure model rather than by adding a separate spatial residual term to the outcome model. The role of ϕ is to regularize the latent exposure surface and borrow strength across neighboring areas.

Under the causal assumptions in Section 2, if the latent true exposure A_i were observed, inference for $\mu(a)$ would proceed by averaging the potential outcome mean under intervention level a over the target covariate distribution. A_i is not observed. Therefore, posterior inference for $\mu(a)$ additionally relies on the assumed measurement error model for $Z_i | A_i, D_i$ and the latent exposure model for $A_i | X_i, \phi_i$. When external calibration data are available, they can be used to inform η_Z through an augmented calibration likelihood or through informative priors. When such calibration information is limited, inference for A_i becomes more dependent on the joint latent exposure and surrogate model assumptions.

Because the covariate distribution $P(L, X)$ is handled nonparametrically, we write the likelihood conditional on the observed covariates (L, X) . The complete-data likelihood, conditional on the fixed spatial structure W , is

$$\begin{aligned} L_c(\eta_Y, \eta_Z, \eta_A, \phi, \psi_\phi; Y, Z, A | L, X, D, W) &= \left\{ \prod_{i=1}^I \prod_{j=1}^{n_i} f_Y(Y_{ij} | A_i, L_{ij}, X_i; \eta_Y) \right\} \left\{ \prod_{i=1}^I f_Z(Z_i | A_i, D_i; \eta_Z) \right\} \\ &\quad \times \left\{ \prod_{i=1}^I f_A(A_i | X_i, \phi_i; \eta_A) \right\} f_\phi(\phi | W; \psi_\phi). \end{aligned} \quad (7)$$

Because the latent true exposures $\{A_i\}_{i=1}^I$ are unobserved, the observed-data likelihood is obtained by integrating over the latent exposures:

$$L_o(\eta_Y, \eta_Z, \eta_A, \phi, \psi_\phi; Y, Z | L, X, D, W) = \int L_c(\eta_Y, \eta_Z, \eta_A, \phi, \psi_\phi; Y, Z, A | L, X, D, W) dA. \quad (8)$$

Let $\pi_Y(\eta_Y)$, $\pi_Z(\eta_Z)$, $\pi_A(\eta_A)$, and $\pi_\psi(\psi_\phi)$ denote prior distributions of the corresponding model parameters. The joint posterior distribution under the augmented-data formulation is

$$\begin{aligned} p(A, \phi, \eta_Y, \eta_Z, \eta_A, \psi_\phi | Y, Z, L, X, D, W) &\propto L_c(\eta_Y, \eta_Z, \eta_A, \phi, \psi_\phi; Y, Z, A | L, X, D, W) \\ &\quad \times \pi_Y(\eta_Y) \pi_Z(\eta_Z) \pi_A(\eta_A) \pi_\psi(\psi_\phi). \end{aligned} \quad (9)$$

Equivalently, $p(A, \Theta | Y, Z, L, X, D, W) \propto L_c(\Theta; Y, Z, A | L, X, D, W) \pi(\Theta)$, where $\Theta = (\eta_Y, \eta_Z, \eta_A, \phi, \psi_\phi)$ and $\pi(\Theta)$ denotes the joint prior over the model parameters. To avoid unnecessary parametric assumptions on the nuisance covariate distribution while keeping the methodological emphasis on the latent exposure and exposure measurement error components, we model the the observed covariates (L_{ij}, X_i) nonparametrically using Bayesian bootstrap²⁶.

4.4 Proposed Bayesian estimation algorithm

Posterior computation proceeds by Markov chain Monte Carlo under the model in (9). Let $N = \sum_{i=1}^I n_i$ denote the total number of individuals, and let M denote the number of retained posterior draws after burn-in. For each draw $m = 1, \dots, M$, we iteratively:

1. sample $(A^{(m)}, \phi^{(m)}, \eta_Y^{(m)}, \eta_Z^{(m)}, \eta_A^{(m)}, \psi_\phi^{(m)})$ from the joint posterior $p(A, \phi, \eta_Y, \eta_Z, \eta_A, \psi_\phi \mid Y, Z, L, X, D, W)$;
2. draw Bayesian bootstrap weights $\{p_{ij}^{(m)} : i = 1, \dots, I, j = 1, \dots, n_i\} \sim \text{Dirichlet}(1, \dots, 1)$ over the $N = \sum_{i=1}^I n_i$ observed covariate values $\{(L_{ij}, X_i) : i = 1, \dots, I, j = 1, \dots, n_i\}$;
3. for each intervention level a of interest, compute the conditional outcome mean for each observed covariate value,

$$\hat{\mu}_{ij}^{(m)}(a) = \mathbb{E}\left\{Y_{ij} \mid A_i = a, L_{ij}, X_i; \eta_Y^{(m)}\right\}, \quad i = 1, \dots, I, j = 1, \dots, n_i;$$

4. average using the Bayesian bootstrap weights,

$$\mu^{(m)}(a) = \sum_{i=1}^I \sum_{j=1}^{n_i} p_{ij}^{(m)} \hat{\mu}_{ij}^{(m)}(a),$$

where $\sum_{i=1}^I \sum_{j=1}^{n_i} p_{ij}^{(m)} = 1$;

5. for any two exposure levels a and a' , compute the posterior draw of the causal contrast $\Delta^{(m)}(a, a') = \mu^{(m)}(a) - \mu^{(m)}(a')$.

Posterior summaries for the average potential outcome and causal contrast are obtained from the empirical distributions of $\{\mu^{(m)}(a)\}_{m=1}^M$ and $\{\Delta^{(m)}(a, a')\}_{m=1}^M$.

5 Simulation Study

We designed the simulation study to evaluate uncertainty propagation under varying degrees of measurement error and model misspecification. In each scenario, we compared the proposed method with an oracle analysis using the true exposure, a naive analysis using the observed error-prone exposure, and a regression calibration approach via plug in imputation. Performance was assessed in terms of bias, RMSE, interval coverage, and interval width for average potential outcomes and causal contrasts across a grid of intervention levels, under varying degrees of measurement error, spatial dependence, sample size, and model misspecification.

Let $i = 1, \dots, I$ index postal code areas. For each area i , let:

- Y_i denote the observed count outcome (number of health events)
- N_i denote the population size (offset)
- A_i denote the latent true area-level $\text{PM}_{2.5}$ exposure (unobserved)
- Z_i denote the observed error-prone surrogate exposure
- $\mathbf{X}_i = (X_{i1}, X_{i2}, X_{i3}, X_{i4})^\top$ denote area-level confounders
- D_i denote the calibration covariate vector explaining measurement error
- ϕ_i denote the spatial random effect on the exposure model
- v_i denote the spatial random effect on the outcome model

For each area $i = 1, \dots, I$, we generate four independent standard normal area-level confounders:

$$X_{ik} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad k = 1, 2, 3, 4 \quad (10)$$

The calibration covariate, capturing characteristics such as rurality or distance from the nearest monitor, is drawn from:

$$D_i \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1) \quad (11)$$

Area population sizes are drawn from:

$$N_i \stackrel{\text{iid}}{\sim} \text{Uniform}(10, 1000) \quad (12)$$

The latent true area-level exposure A_i is generated from a normal distribution. The latent true area-level exposure A_i is generated from a Normal distribution whose mean depends on area-level confounders and, in the spatial extension, a CAR random effect ϕ_i

$$A_i | \mathbf{X}_i, \phi_i \sim \mathcal{N}(\mu_{A,i} + \phi_i, \sigma_A^2) \quad (13)$$

where

$$\mu_{A,i} = \gamma_0 + 0.5X_{i1} - 0.5X_{i2} - 0.5X_{i3} + 0.5X_{i4} \quad (14)$$

We assign a mean-zero proper conditional autoregressive (CAR) prior for random intercepts of the exposure model $\boldsymbol{\phi} = (\phi_1, \dots, \phi_I)^\top$:

$$\boldsymbol{\phi} | \rho_A, \sigma_\phi^2 \sim \mathcal{N}(\mathbf{0}, \sigma_\phi^2(\mathbf{D}_W - \rho_A \mathbf{W})^{-1}) \quad (15)$$

where $\mathbf{W}$ is the $I \times I$ first-order adjacency matrix, and $\mathbf{D}_W = \text{diag}(\mathbf{W})$ is the diagonal degree matrix containing the number of neighbours for each region, and $\rho_A \in [0, 1)$ controls the strength of spatial dependence. The observed error-prone surrogate Z_i is generated from a normal distribution centred on a linear function of the true exposure A_i and the calibration covariate D_i :

$$Z_i | A_i, D_i \sim \mathcal{N}(\beta_{Z1}A_i + \beta_{Z2}D_i, \sigma_Z^2) \quad (16)$$

- β_{Z1} : scaling of the true exposure in the surrogate.
- β_{Z2} : directional bias introduced by D_i . $\beta_{Z2} > 0$ means areas with higher D_i (e.g. farther from monitors) have systematically overestimated surrogates; $\beta_{Z2} < 0$ implies underestimation.
- σ_Z^2 : variance of the measurement error, controlling the magnitude of noise independently of its direction.

The area-level count outcome Y_i is generated from a Poisson distribution with mean $\mu_i = N_i \cdot \exp(\eta_i)$:

$$Y_i | A_i, \mathbf{X}_i, N_i, \mathbf{v}_i \sim \text{Poisson}(N_i \cdot \exp(\eta_i)) \quad (17)$$

where $\log(N_i)$ enters as an offset and $\mathbf{v}_i$ is a spatial random effect. Three functional forms for η_i are considered: a linear, a quadratic and a non-linear form $\eta_i = f(X_i) + \mathbf{v}_i$. Similar to the exposure model, we have $\mathbf{v} = (\mathbf{v}_1, \dots, \mathbf{v}_I)^\top$ to be a CAR prior, independent of $\boldsymbol{\phi}$ such that

$$\mathbf{v} | \rho_Y, \sigma_v^2 \sim \mathcal{N}(\mathbf{0}, \sigma_v^2(\mathbf{D}_W - \rho_Y \mathbf{W})^{-1}) \quad (18)$$

The true average potential outcome under intervention $A_i = a$ for all areas is identified by the g-formula:

$$\mu(a) = \mathbb{E}[Y_i(a)] = \frac{\sum_{i=1}^I N_i \exp(\eta_i(a))}{\sum_{i=1}^I N_i} \quad (19)$$

where $\eta_i(a)$ denotes η_i evaluated at $A_i = a$ with $\mathbf{X}_i$ held at their observed values. This population-weighted average is computed analytically from the true data-generating mechanism and serves as the benchmark for evaluating bias.

6 Summary of Parameters

Parameter	Default value	Description
μ_0	30	Mean true exposure
σ_A	15	SD of true exposure
β_{Z1}	5	Exposure scaling in surrogate
β_{Z2}	0	Directional bias from D_i
σ_Z	1	Measurement error SD
ρ_A	0	Spatial correlation in exposure
ρ_Y	0	Spatial correlation in outcome
I	100	Number of areas
N_i	Uniform(10, 1000)	Area population size

This project will estimate the causal effects of wildfire smoke $PM_{2.5}$ on area-level health outcomes in older adults using the data from the Canadian Longitudinal Study of Aging (CLSA). The CLSA is a national longitudinal study on aging that provides repeated measurements of health, memory and cognitive function from over 50 000 Canadians aged 45 to 85 years²⁷. In particular, we will estimate the causal effect of baseline cumulative wildfire smoke $PM_{2.5}$ exposure on the individual risk of incident depression at follow-up, using Bayesian g-computation to report interpretable causal contrasts (e.g., risk differences under exposure interventions). The analysis will also consider acute impacts on health system utilization and the effects of repeated exposures over time.

Environmental exposures will be assigned using high-resolution spatiotemporal models linked to residential locations, including wildfire smoke $PM_{2.5}$ estimated from the Canadian Optimized Statistical Smoke Exposure Model (CanOSSEM)⁵. As these exposures are measured at aggregated spatial units, they may not reflect individual-level exposure, introducing measurement error and spatial misalignment. This project addresses this challenge by treating true exposure as a latent variable within a Bayesian causal framework. Using Bayesian g-computation, the proposed approach will estimate counterfactual health outcomes under alternative exposure scenarios while incorporating uncertainty due to exposure error. The analysis will compare estimates from naive models that treat exposure as error-free with those obtained under the proposed framework to assess the impact of measurement error on causal effect estimation.

The resulting estimates will provide insight into how wildfire smoke exposures influence health outcomes in older adults, and how accounting for exposure uncertainty may alter conclusions drawn from environmental health studies. These results aim to support more reliable, decision-relevant evidence for population health and climate adaptation.

7 Limitations

There are several limitations of the proposed framework that should be acknowledged. One such limitation is that despite explicitly modeling exposure measurement error, the analysis remains fundamentally constrained by the use of area-level exposure assignments. Environmental exposures derived from models are linked to residential locations such as postal codes and therefore may not capture true individual-level exposure. The proposed framework partially addresses this by modelling a latent area-level exposure A_i as a function of measured area-level characteristics X_i and spatial structure, thereby allowing the observed surrogate exposure to be treated as error-prone rather than fixed and error-free. The framework should be interpreted as a first step toward incorporating exposure uncertainty, rather than a complete solution to individual-level exposure misclassification.

The validity of causal effect estimates relies heavily on the correct specification of both the outcome model and the exposure measurement error model. In particular, identification depends on assumptions made about the error model and the relationship between the latent exposure and the outcome.

Furthermore, complex spatial structures can introduce bias and challenges for causal estimation. Although spatial dependence may be partially incorporated through area-level covariates or latent spatial effects, more complex phenomena such as spatially structured unmeasured confounding, interference between neighbouring regions, or spillover effects are not explicitly modeled and thus introduce additional bias²⁵. In the Bayesian framework, alternative model specifications can be assessed with different modelling comparative diagnostics such as WAIC or cross-validation, and sensitivity analyses should be employed to assess the impact of misspecified modelling assumptions.

Finally, the proposed analysis assumes a static exposure framework and does not fully capture the dynamic and time-varying nature of environmental exposures. While cumulative exposure measures provide a useful summary, they may obscure temporal patterns that are important for understanding health effects. Extending the framework to longitudinal settings with time-varying exposures introduces additional challenges, since measurement error in time-varying covariates can substantially bias causal estimators²⁸. Future work could explore extensions that address time-varying covariates such as marginal structural models and longitudinal g-computation.

8 Expected Outcomes and Timeline of Thesis

The aim of this work is to develop a Bayesian g-computation framework for estimating causal effects under error-prone environmental exposures. The finding will be used to write and defend an MSc thesis in accordance with the requirements set by the Dalla Lana School of Public Health Division of Biostatistics. We also aimed to publish a methodological manuscript to be submitted as well presenting this work at the 2027 statistical conference. The timeline below provides an approximate research schedule:

Time Line For Thesis Project	Year 1				Year 2			
	Fall	Winter	Spring	Summer	Fall	Winter	Spring	Summer
Course Work	X	X						
Literature Review		X	X					
Method Development & Simulation			X	X	X			
Data Analysis					X			
Writing & First Draft						X		
Revision & Final Draft							X	X
Defense								X

References

- [1] World Health Organization. Air pollution. World Health Organization Website, 2026. URL <https://www.who.int/health-topics/air-pollution>. Accessed: March 15, 2026.
- [2] Zhe Wang, Zifa Wang, Zhiyin Zou, Xueshun Chen, Huangjian Wu, Wending Wang, Hang Su, Fang Li, Wenru Xu, Zhihua Liu, et al. Severe global environmental issues caused by canada's record-breaking wildfires in 2023, 2024.
- [3] Piyush Jain, Quinn E Barber, Stephen W Taylor, Ellen Whitman, Dante Castellanos Acuna, Yan Boulanger, Raphaël D Chavardès, Jack Chen, Peter Englefield, Mike Flannigan, et al. Drivers and impacts of the record-breaking 2023 wildfire season in canada. *Nature Communications*, 15(1): 6764, 2024.
- [4] Carlyn J Matz, Marika Egyed, Guoliang Xi, Jacinthe Racine, Radenko Pavlovic, Robyn Rittmaster, Sarah B Henderson, and David M Stieb. Health impact analysis of pm_{2.5} from wildfire smoke in canada (2013–2015, 2017–2018). *Science of The Total Environment*, 725:138506, 2020.
- [5] Naman Paul, Jiayun Yao, Kathleen E. McLean, David M. Stieb, and Sarah B. Henderson. The canadian optimized statistical smoke exposure model (canossem): A machine learning approach to estimate national daily fine particulate matter (pm_{2.5}) exposure. *Science of the Total Environment*, 850:157956, 2022. <https://doi.org/10.1016/j.scitotenv.2022.157956>.
- [6] Jeffrey R Brook, Eleanor M Setton, Evan Seed, Mahdi Shooshtari, and Dany Doiron. The canadian urban environmental health research consortium—a protocol for building a national environmental exposure data platform for integrated analyses of urban form and health. *BMC public health*, 18(1):114, 2018.
- [7] Alexandros Gryparis, Christopher J Paciorek, Ariana Zeka, Joel Schwartz, and Brent A Coull. Measurement error caused by spatial misalignment in environmental epidemiology. *Biostatistics*, 10(2):258–274, 2009.
- [8] Jennifer Richmond-Bryant and Thomas C Long. Influence of exposure measurement errors on results from epidemiologic studies of different designs. *Journal of Exposure Science & Environmental Epidemiology*, 30(3):420–429, 2020.
- [9] Yaguang Wei, Xinye Qiu, Mahdieh Danesh Yazdi, Alexandra Shtein, Liuhua Shi, Jiabei Yang, Adjani A Peralta, Brent A Coull, and Joel D Schwartz. The impact of exposure measurement error on the estimated concentration–response relationship between long-term exposure to pm_{2.5} and mortality. *Environmental health perspectives*, 130(7):077006, 2022.
- [10] Evangelia Samoli, Barbara K Butland, Sophia Rodopoulou, Richard W Atkinson, Benjamin Barratt, Sean D Beevers, Andrew Beddows, Konstantina Dimakopoulou, Joel D Schwartz, Mahdieh Danesh Yazdi, et al. The impact of measurement error in modeled ambient particles exposures on health effect estimates in multilevel analysis: a simulation study. *Environmental Epidemiology*, 4(3):e094, 2020.
- [11] Dimitris Evangelopoulos, Dylan Wood, Ben Barratt, Hanbin Zhang, Audrey De Nazelle, Sean Beevers, Barbara K Butland, Evangelia Samoli, Joel Schwartz, Kees de Hoogh, et al. Exposure measurement error in air-pollution epidemiology and its determinants: results from the melons study. *International Journal of Epidemiology*, 55(1):dyaf214, 2026.
- [12] Adam Szpiro, Lianne Sheppard, and Thomas Lumley. Measurement error in air pollution cohort studies. *Epidemiology*, 22(1):S32, 2011.

- [13] Mariam S Girguis, Lianfa Li, Fred Lurmann, Jun Wu, Carrie Breton, Frank Gilliland, Daniel Stram, and Rima Habre. Exposure measurement error in air pollution studies: the impact of shared, multiplicative measurement error on epidemiological health risk estimates. *Air Quality, Atmosphere & Health*, 13(6):631–643, 2020.
- [14] Thomas F. Bateson and J. Michael Wright. Regression calibration for classical exposure measurement error in environmental epidemiology studies using multiple local surrogate exposures. *American Journal of Epidemiology*, 172(3):344–352, 2010. <https://doi.org/10.1093/aje/kwq123>.
- [15] Stacey E Alexeeff, Raymond J Carroll, and Brent Coull. Spatial measurement error and correction by spatial simex in linear regression models when using predicted air pollution exposures. *Biostatistics*, 17(2):377–389, 2016.
- [16] Changwoo J Lee, Elaine Symanski, Amal Rammah, Dong Hun Kang, Philip K Hopke, and Eun Sug Park. A scalable two-stage bayesian approach accounting for exposure measurement error in environmental epidemiology. *Biostatistics*, 26(1):kxae038, 2025.
- [17] Saskia Comess, Howard H Chang, and Joshua L Warren. A bayesian framework for incorporating exposure uncertainty into health analyses with application to air pollution and stillbirth. *Biostatistics*, 25(1):20–39, 2024.
- [18] Alexander P Keil, Eric J Daza, Stephanie M Engel, Jessie P Buckley, and Jessie K Edwards. A bayesian approach to the g-formula. *Statistical methods in medical research*, 27(10):3183–3204, 2018.
- [19] Alexander P Keil, Jessie P Buckley, and Amy E Kalkbrenner. Bayesian g-computation for estimating impacts of interventions on exposure mixtures: demonstration with metals from coal-fired power plants and birth weight. *American journal of epidemiology*, 190(12):2647–2657, 2021.
- [20] Xiao Wu, Danielle Braun, Marianthi-Anna Kioumourtoglou, Christine Choirat, Qian Di, and Francesca Dominici. Causal inference in the context of an error prone exposure: air pollution and mortality. *The annals of applied statistics*, 13(1):520, 2019.
- [21] Danielle Braun, Malka Gorfine, Giovanni Parmigiani, Nils D Arvold, Francesca Dominici, and Corwin Zigler. Propensity scores with misclassified treatment assignment: a likelihood-based adjustment. *Biostatistics*, 18(4):695–710, 2017. <https://doi.org/10.1093/biostatistics/kxx014>.
- [22] Joel Schwartz, Yuzhou Feng, Eric Castro, and Yaguang Wei. Causal concentration-response modeling with continuous curves and exposure error correction: PM_{2.5} and mortality in the medicare cohort. *Environmental Health Perspectives*, 133(6):067007, 2025. <https://doi.org/10.1289/EHP15238>.
- [23] Kevin P. Josey, Priyanka deSouza, Xiao Wu, Danielle Braun, and Rachel Nethery. Estimating a causal exposure response function with a continuous error-prone exposure: A study of fine particulate matter and all-cause mortality. *Journal of Agricultural, Biological and Environmental Statistics*, 28(1):20–41, 2023. <https://doi.org/10.1007/s13253-022-00508-z>.
- [24] Georgia Papadogeorgou, Christine Choirat, and Corwin M Zigler. Adjusting for unmeasured spatial confounding with distance adjusted propensity score matching. *Biostatistics*, 20(2):256–272, 2019. <https://doi.org/10.1093/biostatistics/kxx074>.
- [25] Georgia Papadogeorgou and Francesca Dominici. A causal exposure response function with local adjustment for confounding: estimating health effects of exposure to low levels of ambient fine particulate matter. *The Annals of Applied Statistics*, 14(2):850–871, 2020. <https://doi.org/10.1214/20-AOAS1330>.

- [26] Arman Oganisian and Jason A. Roy. A practical introduction to bayesian estimation of causal effects: Parametric and nonparametric approaches. *Statistics in Medicine*, 40(2):518–551, 2021. <https://doi.org/10.1002/sim.8761>.
- [27] Parminder S Raina, Christina Wolfson, Susan A Kirkland, Lauren E Griffith, Mark Oremus, Christopher Patterson, Holly Tuokko, Margaret Penning, Cynthia M Balion, David Hogan, et al. The canadian longitudinal study on aging (clsa). *Canadian Journal on Aging/La Revue canadienne du vieillissement*, 28(3):221–229, 2009.
- [28] Ryan P Kyle, Erica EM Moodie, Marina B Klein, and Michał Abrahamowicz. Correcting for measurement error in time-varying covariates in marginal structural models. *American journal of epidemiology*, 184(3):249–258, 2016.